Module 7: Classification

Part 3: naive Byes

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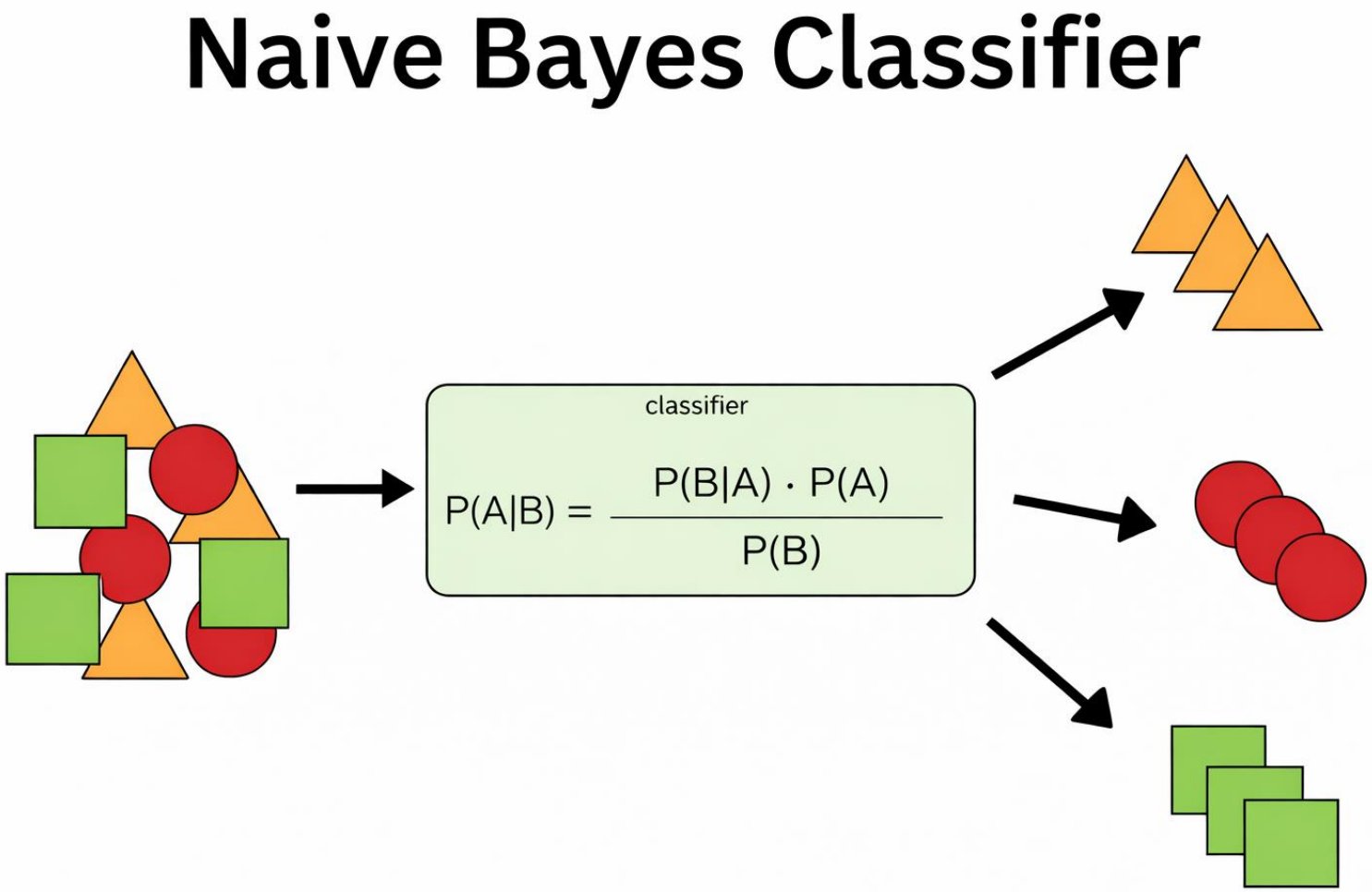
Some lecture slides and instructional materials in this course are adapted from the following sources:
https://statquest.org/index.html

Naive Bayes



Naive Bayes

Recall that in LDA/QDA, we used Bayes Theorem to approximate $P(Y = k \mid X)$

$$P(Y = k \mid X) = \frac{P(X \mid Y = k)P(Y = k)}{P(X)}$$

- $\boldsymbol{P(X \mid Y = k) = f_k(x)}$ is the p-dimensional density function for an observation in the k th class.
- Estimating this density function is VERY challenging.
- LDA/QDA makes strong assumptions about this density function. This makes a very challenging problem much simpler.
- Naive Bayes makes a different attempt to estimate $P(X \mid Y = k)$.

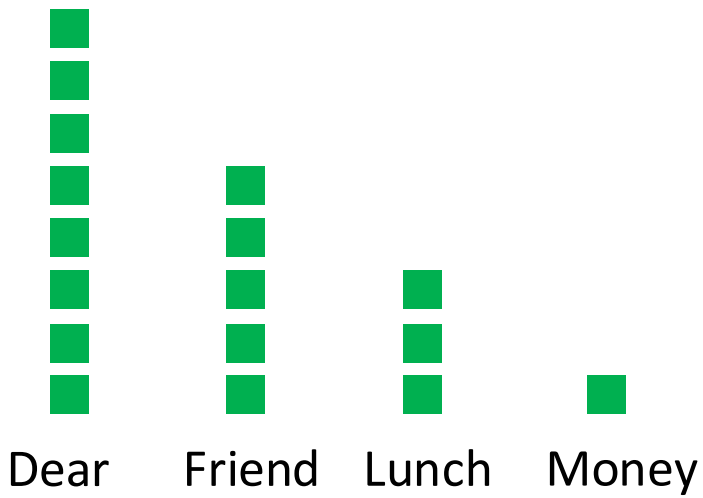
Naive Bayes' Assumption

- Within the k th class, the p predictors are independent.
- Recall, we denote $P(X \mid Y = k) = f_k(x)$ where $x = (x_1, x_2, \dots, x_p)$.

Mathematically, this means for $k = 1, \dots, K$:

$$f_k(x_1, x_2, x_3, \dots, x_p) = f_{k1}(x_1) \times f_{k1}(x_2) \times \cdots \times f_{kp}(x_p)$$

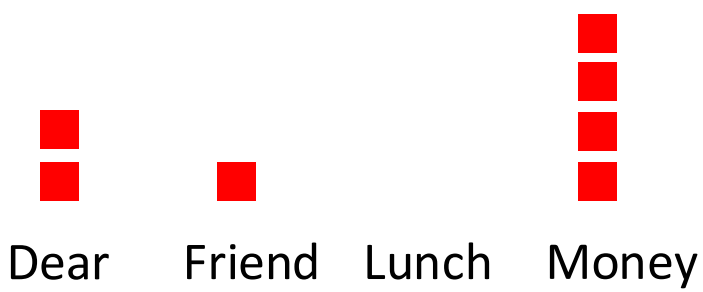
Spam Email Classifier



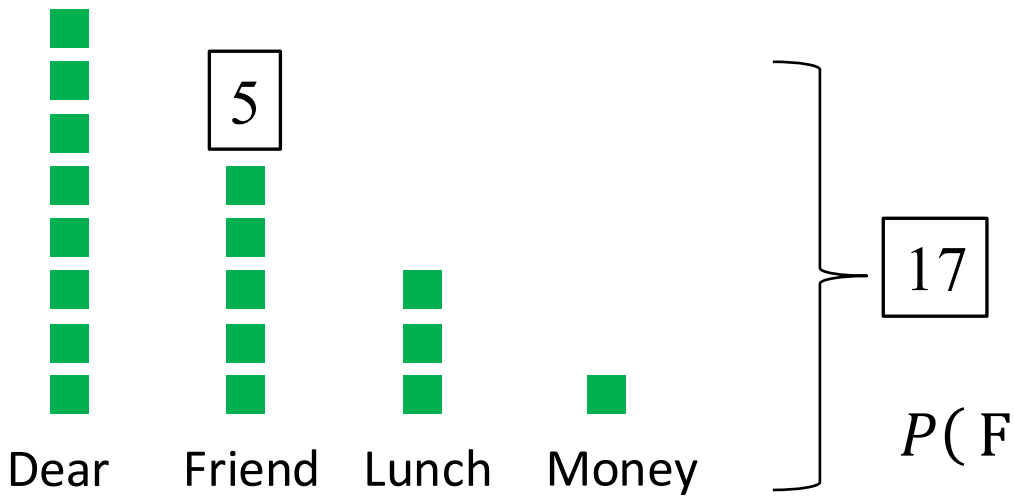
Prior Probability (Initial guess)

$$P(\text{Normal}) = \frac{8}{8+4} = 0.67$$

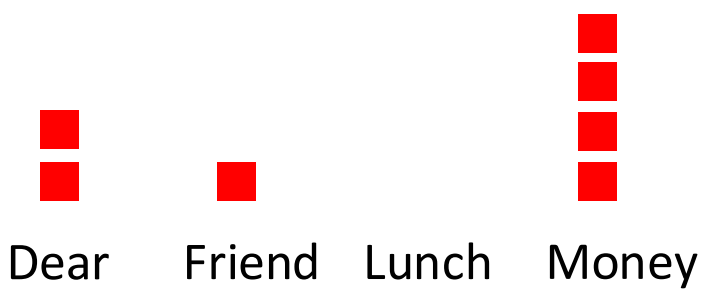
$$P(\text{Spam}) = \frac{4}{8+4} = 0.33$$



Spam Email Classifier



$$P(\text{Friend} \mid \text{Normal}) = \frac{5}{17} = 0.29$$



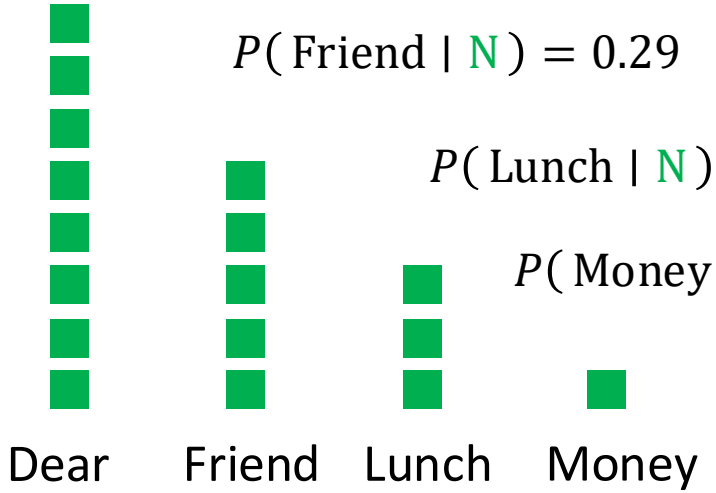
Spam Email Classifier

$P(\text{Dear} \mid \text{N}) = 0.47$

$P(\text{Friend} \mid \text{N}) = 0.29$

$P(\text{Lunch} \mid \text{N}) = 0.18$

$P(\text{Money} \mid \text{N}) = 0.06$

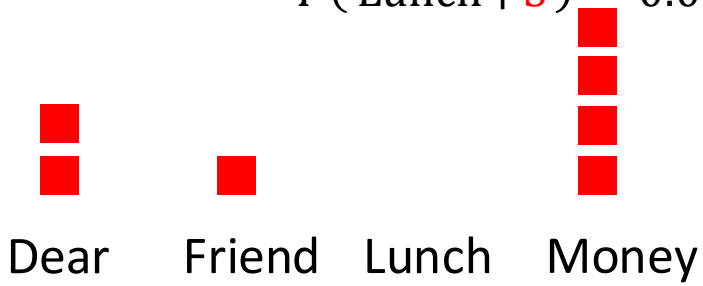


$\text{P}(\text{Dear} \mid \text{S}) = 0.29$

$P(\text{Friend} \mid \text{S}) = 0.14$

$P(\text{Lunch} \mid \text{S}) = 0.00$

$P(\text{Money} \mid \text{S}) = 0.57$



Spam Email Classifier

Email: Dear Friend

$P(\text{Dear} \mid \text{N}) = 0.47$
$P(\text{Friend} \mid \text{N}) = 0.29$
$P(\text{Lunch} \mid \text{N}) = 0.18$
$P(\text{Money} \mid \text{N}) = 0.06$

$$P(N) \times P(Dear \mid N) \times P(Friend \mid N)$$

$$=$$

$P(\text{Dear} \mid \text{S}) = 0.29$
$P(\text{Friend} \mid \text{S}) = 0.14$
$P(\text{Lunch} \mid \text{S}) = 0.00$
$P(\text{Money} \mid \text{S}) = 0.57$

$$P(S) \times P(Dear \mid S) \times P(Friend \mid S)$$

$$=$$

Spam Email Classifier

$P(\text{Normal}) = 0.67$

$P(\text{Dear} \mid \text{N}) = 0.47$
$P(\text{Friend} \mid \text{N}) = 0.29$
$P(\text{Lunch} \mid \text{N}) = 0.18$
$P(\text{Money} \mid \text{N}) = 0.06$

Email: Lunch Money Money Money Money

$$P(N) \times P(Money \mid N) \times P(Money \mid N) \times P(Money \mid N) \times P(Money \mid N)$$

$=$

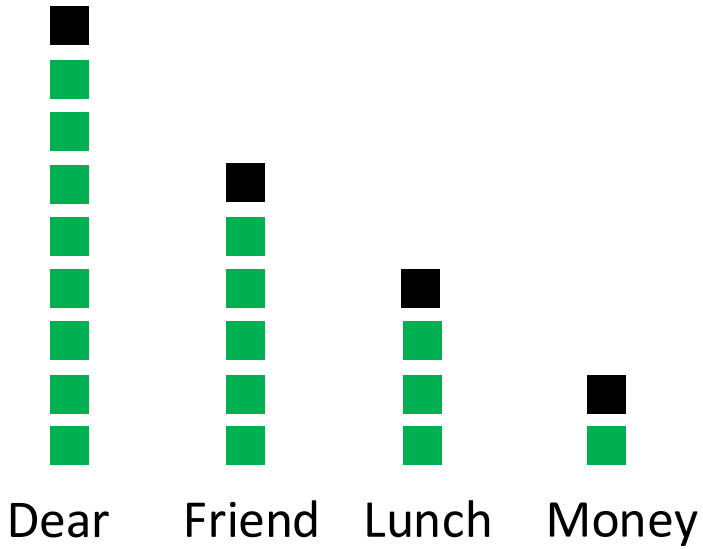
$P(\text{Spam}) = 0.33$

$P(\text{Dear} \mid \text{S}) = 0.29$
$P(\text{Friend} \mid \text{S}) = 0.14$
$P(\text{Lunch} \mid \text{S}) = 0.00$
$P(\text{Money} \mid \text{S}) = 0.57$

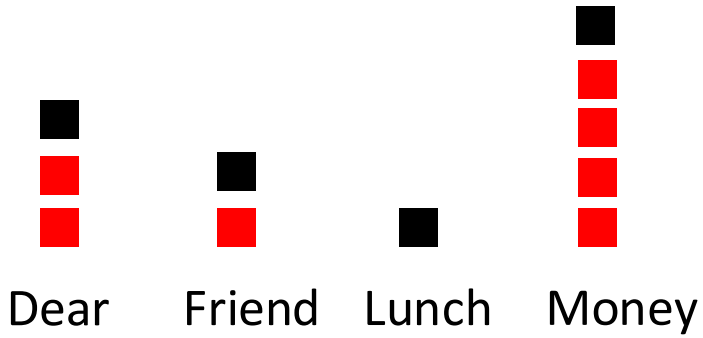
$$P(S) \times P(Money \mid S) \times P(Money \mid S) \times P(Money \mid S) \times P(Money \mid S)$$

$=$

Spam Email Classifier



Black box: Add 1 count



Spam Email Classifier

$P(\text{Normal}) = 0.67$

$P(\text{Dear} \mid \text{N}) = 0.43$
$P(\text{Friend} \mid \text{N}) = 0.29$
$P(\text{Lunch} \mid \text{N}) = 0.19$
$P(\text{Money} \mid \text{N}) = 0.10$

Email: Lunch Money Money Money Money

$$P(N) \times P(Money \mid N) \times P(Money \mid N) \times P(Money \mid N) \times P(Money \mid N)$$

$=$

$P(\text{Spam}) = 0.33$

$P(\text{Dear} \mid \text{S}) = 0.27$
$P(\text{Friend} \mid \text{S}) = 0.18$
$P(\text{Lunch} \mid \text{S}) = 0.09$
$P(\text{Money} \mid \text{S}) = 0.45$

$$P(S) \times P(Money \mid S) \times P(Money \mid S) \times P(Money \mid S) \times P(Money \mid S)$$

$=$

Spam Email Classifier

Assumption

- All words are independent
- Ignore relationshipes between words

Naive Bayes' Assumption

- Do we really believe this assumption? No.
- Even though this is a highly naive assumption, it often leads to relatively decent results.
- From the viewpoint of the bias-variance tradeoff: Naive Bayes introduces some bias, but reduces variance, leading to a classifier that actually works quite well in practice.

What does the problem boil down to?

$$P(Y = k \mid X) = \frac{P(X \mid Y = k)P(Y = k)}{P(X)}$$

$$= \frac{f_k(X)\pi_k}{P(X)}$$

$$= \frac{f_{k1}(x_1) \times f_{k1}(x_2) \times \cdots \times f_{kp}(x_p)\pi_k}{P(X)}$$

Options for estimating these marginal densities $\boldsymbol{f_{kj}}$:

- If the predictor is **qualitative**, count the proportion of trainings observations for the predictor corresponding to each class.
- If the predictor is **quantitative**, use a parametric approach (if we know its distribution) or a non-parametric approach (kernel density estimation).

Naive Bayes

- Trades one strong assumption for another.
- If you have qualitative predictors, Naive Bayes is a good choice (because normality assumption definitely does not hold).
- When n is not large enough relative to p, Naive Bayes can also outperform LDA/QDA.

Implementation

- See R script: naivebayes.R